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United States Patent	12404846
Kind Code	B2
Date of Patent	September 02, 2025
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Electronic pressure compensated hydraulic motor pump with variable output power

Abstract

An electronic pressure compensated hydraulic motor pump is controlled to provide variable output power based on a variable input signal. The variable output power feature allows the motor pump to be used with a power management system to better match the output power of the motor pump with the available power of an electrical system. The ability to provide variable output power provides beneficial power management for electrical systems that switch between different power modes, e.g., between a generator power mode and a battery power mode.

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Appl. No.: 17/933978

Filed: September 21, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230102332 A1	Mar. 30, 2023

Related U.S. Application Data

us-provisional-application US 63246366 20210921

Publication Classification

Int. Cl.: **F04B17/03** (20060101); **F04B49/06** (20060101); **F04B49/20** (20060101); **H02P4/00** (20060101); **H02P6/06** (20060101); F04B1/20 (20200101)

U.S. Cl.:

CPC **F04B17/03** (20130101); **F04B49/065** (20130101); **F04B49/20** (20130101); **H02P4/00** (20130101); **H02P6/06** (20130101); F04B1/20 (20130101); F04B2201/1201 (20130101); F04B2203/0201 (20130101); F04B2203/0202 (20130101); F04B2203/0208 (20130101); F04B2203/0209 (20130101); F04B2205/05 (20130101); F04B2205/06 (20130101); F04B2205/09 (20130101); Y02T50/80 (20130101)

Field of Classification Search

CPC: F04B (1/20); F04B (17/03); F04B (49/065); F04B (49/20); F04B (2201/1201); F04B (2203/0201); F04B (2203/0202); F04B (2203/0208); F04B (2203/0209); F04B (2205/05); F04B (2205/06); F04B (2205/09); H02P (4/00); H02P (6/06)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/246,366, filed Sep. 21, 2021, and titled “Electronic Pressure Compensated Hydraulic Motor Pump with Variable Output Power,” the disclosure of which is hereby incorporated by reference.

BACKGROUND

(1) The commercial aircraft industry is currently evaluating means to significantly reduce or eliminate CO₂ emissions with the next generation aircraft and support equipment. Government initiatives such as EU's Clean Sky 2025-2050 is establishing targets and timing for aircraft OEMs to reduce CO₂ emissions. More electric aircraft utilizing battery fuel cells is one possible solution toward achieving reduced emissions goals. Elimination of fossil fuel ground support equipment is an additional CO₂ focus for reduction during gate transitions. Today ground carts convert fossil fuel to power the aircraft's electric motor driven hydraulic pumps used to open/close cargo doors, verify function of flight control actuators, and perform maintenance operations during gate transitions. The aircraft's hydraulic pumps are fixed pressure, sized to power flight control actuators subjected to high loads during flight. During gate operation the hydraulic pump consumes the same power as during flight which is not required during ground operations.

SUMMARY

(2) An electronic pressure compensated hydraulic motor pump, according to the present disclosure, provides a variable output power electric motor driven hydraulic pump that reduces consumed power during gate transitions of an aircraft while providing the full power needed by the aircraft during flight. The electronic pressure compensated hydraulic motor pump enables an aircraft to manage the battery fuel cell life by using only the power needed to perform an intended function without demanding excessive un-needed power. This capability assists in eliminating a ground service cart and supports optimizing the battery size reducing mean take-off weight thus reducing fuel consumption during flight. More generally, the electronic pressure compensated hydraulic motor pump is controlled to provide variable output power based on a variable input signal. The variable output power feature allows the motor pump to be used with a power management system to better match the output power of the motor pump with the available power of an electrical system. The ability to provide variable output power provides beneficial power management for electrical systems that switch between different power modes, e.g., between a generator power mode and a battery power mode.

(3) An aspect of the present disclosure relates to a fluid device system. The fluid device system includes a fluid pump, an electric motor and a controller. The fluid pump has a fixed displacement and includes a fluid inlet and a fluid outlet. The electric motor has a shaft coupled to the fluid pump and the shaft rotates in response to a power command signal received at the electric motor. The controller receives a first input value and produces an output to communicate the power command signal to the electric motor. The controller adjusts the power command signal to operate the electrical motor in a normal operating mode and a input control operating mode. In the normal operating mode, the controller refers to a first lookup table using the first input value and outputs a first power command signal to the electrical motor based on the referral to the first lookup table. In the input control mode, the controller determines a second power control signal such that the fluid pump operated at a reduced capacity in comparison to the normal operating mode. In certain embodiments, a scalar value is used by the controller to determine the second power control signal.

(4) Another aspect of the present disclosure relates to a method of controlling a fluid device system having a fixed displacement fluid pump, an electric motor having a rotational shaft coupled to the fluid pump, and a controller including an output to communicate a command signal to the electric motor. The method includes: (a) receiving at the controller an input directing operation in either a normal operating mode or an input control operating mode; (b) in response to an input directing operation in a normal mode, receiving a first input value at the controller, using the first input value to reference a first output value in a lookup table of the controller, and outputting from the controller a first power command signal that corresponds to the first output value to the electrical motor; and (c) in response to an input directing operation in an input control operating mode, receiving a scalar value at the controller, scaling at the controller at least a portion of the content of

the lookup table with the scalar value, receiving a second input value at the controller, using the second input value to reference a second output value in the scaled lookup table, and outputting from the controller a second power command signal that corresponds to the second output value to the electrical motor.

(5) A variety of additional aspects will be set forth in the description that follows. These aspects can relate to individual features and to combinations of features. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the broad concepts upon which the embodiments disclosed herein are based.

Description

DRAWINGS

(1) FIG. 1 is a schematic representation of a simplified hydraulic system that includes an electronic pressure compensated hydraulic motor pump according to the present disclosure.

(2) FIG. 2 is a cross-sectional view of an example of a rotary fluid device suitable for use with system of FIG. 1.

(3) FIG. 3 is a cross-sectional view of the rotary fluid device taken on line 3-3 of FIG. 2.

(4) FIG. 4 is a schematic representation of a controller suitable for use with the system of FIG. 1.

(5) FIG. 5 is a pump curve of electronic pressure compensated hydraulic motor pump whose performance is 100% of rated output power.

(6) FIG. 6 is a pump curve of an electronic pressure compensated hydraulic motor pump whose performance is 50% of rated output power.

DETAILED DESCRIPTION

(7) An electronic pressure compensated hydraulic motor pump, according to the present disclosure, provides a variable output power electric motor driven hydraulic pump that reduces consumed power during gate transitions of an aircraft while providing the full power needed by the aircraft during flight. The electronic pressure compensated hydraulic motor pump enables an aircraft to manage the battery fuel cell life by using only the power needed to perform an intended function without demanding excessive un-needed power. This capability assists in eliminating a ground service cart and supports optimizing the battery size reducing mean take-off weight thus reducing fuel consumption during flight. More generally, the electronic pressure compensated hydraulic motor pump is controlled to provide variable output power based on a variable input signal. The variable output power feature allows the motor pump to be used with a power management system to better match the output power of the motor pump with the available power of an electrical system. The ability to provide variable output power provides beneficial power management for electrical systems that switch between different power modes, e.g., between a generator power mode and a battery power mode.

(8) Referring now to FIG. 1, a schematic representation of a simplified exemplary hydraulic system, generally designated **10**, is shown. The hydraulic system **10** includes a fluid device system, generally designated **12**, in fluid communication with a fluid reservoir **14** and an actuator **16** (e.g., motor, cylinder, etc.). The fluid device system **12** includes a rotary fluid device, generally designated **18** and a controller, generally designated **20**.

(9) The rotary fluid device **18** includes a fluid pump **22**, an electric motor **24** and a pressure transducer **27** at an outlet of the fluid pump **22**. The fluid pump **22** is a fixed displacement pump that is in engagement with or coupled to the electric motor **24**, which in the illustrated embodiment is a permanent magnet synchronous motor.

(10) In the depicted embodiment of FIG. 1, the fluid pump **22** is in fluid communication with the fluid reservoir **14** and the actuator **16**. While fluid pump **22** is shown in direct fluid communication

with the fluid reservoir **14** and the actuator **16**, it will be understood that the scope of the present disclosure is not limited to the fluid pump **22** being in direct fluid communication with the fluid reservoir **14** and actuator **16** as any number of valves or other fluid components could be disposed between the fluid pump **22** and the fluid reservoir **14** and/or actuator **16**.

(11) In the depicted embodiment, the electric motor **24** is in electrical communication with the controller **20**. As will be described in greater detail herein, the controller **20** outputs an electrical signal **25** to the electric motor **24**. In response to the electrical signal **25**, a shaft **26** of the electric motor **24** rotates. As the fluid pump **22** is a fixed displacement pump and as the fluid pump **22** is in engagement with the shaft **26** of the electric motor **24**, the rotation of the shaft **26** causes the fluid pump **22** to transfer fluid from the fluid reservoir **14** to the actuator **16**.

(12) Referring to FIG. 2, an example configuration of a rotary fluid device **18** is shown; other suitable rotary fluid devices may also be utilized in the hydraulic system **10** and/or in conjunction with the controller **20**. The exemplary rotary fluid device **18** includes a housing, generally designated **28**. The housing **28** includes a fluid inlet **30** and a fluid outlet **32**. The housing **28** further includes a main body, generally designated **34**, which includes a first end portion **36** and an opposite second end portion **38**, a first end assembly, generally designated **40**, which is adapted for engagement with the first end portion **36** of the main body **34**, and a second end assembly, generally designated **42**, which is adapted for engagement with the second end portion **38**.

(13) The first end portion **36** of the main body **34** defines a first chamber **44** having a first opening **46** while the second end portion **38** defines a second chamber **48** having a second opening **50**. In the depicted embodiment, the first and second openings **46**, **50** are oppositely disposed along a longitudinal axis **52** of the main body **34**. A chamber **54** through the main body **34** connects the first chamber **44** to the second chamber **48**.

(14) In the depicted embodiment, the first chamber **44** is adapted to receive the fluid pump **22** through the first opening **46** while the second chamber **48** is adapted to receive the electric motor **24** through the second opening **50**. The shaft **26** of the electric motor **24** extends through the chamber **54** and is engaged with the fluid pump **22**.

(15) A pumping assembly, generally designated **56**, is disposed in the first chamber **44** of the main body **34**. While the pumping assembly **56** is shown as an axial piston assembly, it will be understood that the pump assembly **56** is not so necessarily limited as the pumping assembly **56** could be a vane assembly, gerotor assembly, cam lobe assembly, etc. In the depicted embodiment, the pumping assembly **56** includes a barrel assembly **58** and an angle block **60**.

(16) The barrel assembly **58** includes a cylinder barrel **62** defining an inner bore. In the depicted embodiment, the inner bore of the cylinder barrel **62** includes a plurality of internal teeth that are adapted for engagement with the shaft **26**.

(17) The cylinder barrel **62** further defines a plurality of axially oriented cylinder bores **64**. Disposed within each cylinder bore **64** is an axially reciprocal piston **66**, which includes a generally spherical head that is pivotally received by a slipper member **68**. The slipper members **68** slide along an inclined surface of the stationary angle block **60**.

(18) The cylinder bores **64** and the pistons **66** cooperatively define a plurality of volume chambers **70**. In response to rotation of the shaft **26**, the cylinder barrel **62** rotates about a rotating axis causing the plurality of volume chambers **70** to expand and contract. In the depicted embodiment, the rotating axis is generally aligned with the longitudinal axis **52**. During rotation of the cylinder barrel **62**, fluid from a fluid source (e.g., the fluid reservoir **14**) is drawn into the expanding volume chambers **70** while fluid from the contracting volume chambers **70** is expelled to a fluid destination (e.g., the actuator **16**).

(19) The first end assembly **40** is engaged with the first end portion **36** of the main body **134**. The first end assembly **40** includes a valving portion **72** having an inlet passage **74** and an outlet passage **76** (shown in FIG. 3). In the depicted embodiment, the inlet and outlet passages **74**, **76** are arcuately shaped fluid passages. The inlet and outlet passages **74**, **76** are adapted for commutating

fluid communication with the volume chambers **70** of the barrel assembly **58**. The expanding volume chambers **70** are in fluid communication with the inlet passage **74** while the contracting volume chambers **70** are in fluid communication with the outlet passage **76**. The inlet passage **74** is in fluid communication with the fluid inlet **30** while the outlet passage **76** is in fluid communication with the fluid outlet **32**. In the depicted embodiment, the fluid outlet **32** is defined by the first end assembly **40**.

(20) The electric motor **24** is disposed in the second chamber **48** of the main body **34**. The electric motor **24** is a three-phase permanent magnet synchronous motor. It will be understood, however, that the scope of the present disclosure is not limited to the electric motor **24** being a three-phase permanent magnet synchronous motor. The electric motor **24** includes a rotor **80** and a stator **82**.

(21) The rotor **80** includes permanent magnets **84** engaged with the shaft **26**. In one embodiment, the permanent magnets **84** are keyed to the shaft **26** so that the permanent magnets **84** rotate with the shaft **26**.

(22) The stator **82** is engaged with the second end portion **38** of the main body **34**. The stator **82** includes a plurality of coils that create an electromagnetic field when current passes through the coils. By energizing the coils of the stator **82**, the permanent magnets **84** rotate causing the shaft **26** to rotate as well.

(23) The second end assembly **42** is engaged with the second end portion **38** of the main body **34**. In the depicted embodiment, the second end assembly **42** includes a plate assembly **88** and a cover assembly **90**.

(24) The plate assembly **88** is engaged with the second opening **50** of the second end portion **38** of the main body **34**. The plate assembly **88** defines a central chamber **92** and a plurality of flow passages **94** (shown in FIG. 3). The central chamber **92** is adapted to receive an end portion **96** of the shaft **26**. In the depicted embodiment, a conventional bearing assembly **98** is engaged in the central chamber **92** such that an inner race of the bearing assembly **98** is in tight-fit engagement with the shaft **26** while an outer race of the bearing assembly **98** is in tight-fit engagement with the central chamber **92**.

(25) The cover assembly **90** defines the fluid inlet **30** for the rotary fluid device **18**. In the depicted embodiment, the cover assembly **90** and the plate assembly cooperatively define a third chamber **100** of the rotary fluid device **18**.

(26) One or more sensors **102** can be disposed in the third chamber **100**. The one or more sensors **102** can include a speed sensor **102a**, a position sensor **102b**, and/or a fluid temperature sensor **102c**. In the depicted embodiment, a conventional resolver is used for the speed sensor **102a** and the position sensor **102b**. The resolver includes a stator portion and a rotor portion. The stator portion includes a plurality of wire windings through which current flows. As the rotor portion rotates, the relative magnitudes of voltages through the wire windings are measured and used to determine speed and position of the rotor portion. In the depicted embodiment, the rotor portion is disposed on the end portion **96** of the shaft **26**.

(27) The fluid temperature sensor **102c** can be used to measure the temperature of the fluid in the rotary fluid device **18**. In the depicted embodiment, the fluid temperature sensor **102c** is engaged with the plate assembly **88** and disposed adjacent to one of the plurality of flow passages **94**. The fluid temperature sensor **102c** can be a conventional resistance temperature detector (RTD). The RTD includes a resistor that changes resistance value as its temperature changes.

(28) Referring now to FIGS. 2 and 3, the flow of fluid through the rotary fluid device **18** will be described. As the shaft **26** of the electric motor **24** rotates, fluid enters the fluid inlet **30** of the second end assembly **42**. The fluid enters the third chamber **100** and passes through the flow passages **94** in the plate assembly **88**. The fluid then enters the second chamber **48** of the main body **34**. In the second chamber **48**, the fluid is in contact with the electric motor **24**. This fluid contact is potentially advantageous as it provides lubrication to the electric motor **24**.

(29) The fluid passes from the second chamber **48** to the first chamber **44** through a fluid pathway

104. The fluid pathway **104** is in fluid communication with the inlet passage **74**. The fluid then enters the expanding volume chamber **70**. As the barrel assembly **58** rotates about the rotating axis, the pistons **66** axially extend and retract from the cylinder bores **64**. As the pistons **66** extend, the volume chambers **70** expand thereby drawing fluid from the inlet passage **74** into the expanding volume chambers. As the pistons **66** contract, the volume chambers **70** contract thereby expelling fluid from the contracting volume chambers **70** through the outlet passage **76** and through the fluid outlet **32**.

(30) Referring now to FIG. **4**, the schematic representation of the simplified exemplary hydraulic system **10** is illustrated once again including a schematic representation of the controller **20**. The controller **20** supplies an electrical signal **25** to the electric motor **24** in order to obtain a desired characteristic (e.g., constant horsepower, pressure compensation, variable pressure compensation, etc.) from the rotary fluid device **18**. The controller **20** uses one or more control algorithms and/or predefined performance data for the electric motor **24** and the fluid pump **22** to control or regulate the rotary fluid device **18**. Through the use of the one or more control algorithms and/or the predefined performance data, the rotary fluid device **18** can be controlled to have constant horsepower characteristics, pressure compensation characteristics, and/or variable pressure compensation characteristics.

(31) In the depicted embodiment, the controller **20** converts a direct current voltage input to an alternating phase current output, which is supplied to the electric motor **24** for driving the pumping assembly **56**. The controller **20** includes a plurality of inputs **110**. In the depicted embodiment, and by way of example only, the plurality of inputs **110** include a voltage input **110a** provided by the power supply **111**, a shaft position input **110b** provided by a resolver **120** operating as both the speed sensor **102a** and position sensor **102b** (see FIG. **2**), a pump output pressure input **110c** provided by the pressure transducer **27**, a normal mode input **110d** provided by an electronic switch **113**, a control mode input **110e** provided by the electronic switch **113**, and an input control port **110f** that provides a voltage or current input to the controller **20**.

(32) The controller **20** further includes a plurality of outputs **112** including a voltage output **112a**, a phase current output **112b** and a phase angle output **112c**. In the depicted embodiment, each of the plurality of outputs **112** is in electrical communication with the electric motor **24**.

(33) The controller **20** further includes a circuit **114** having a microprocessor **116**, a resolver speed calculation Integrated Circuit (IC) **122**, a proportional scalar IC **124**, a Proportional-Integral-Derivative (PID) IC **126**, and a servo drive **127**.

(34) In the depicted embodiment, the microprocessor **116** is a field programmable gate array (FPGA). The FPGA **116** is a semiconductor device having programmable logic components, such as logic gates (e.g., AND, OR, NOT, XOR, etc.) or more complex combinational functions (e.g., decoders, mathematical functions, etc.), and programmable interconnects, which allow the logic blocks to be interconnected. In the depicted embodiment, the FPGA **116** is programmed to provide voltage and current to the electric motor **24** of the rotary fluid device **18** such that the rotary fluid device **18** responds in accordance with desired performance characteristics (e.g., constant horsepower, pressure compensation, constant speed, variable pressure compensation, etc.). Storage media, which can be independent of or incorporated into the microprocessor **116**, the resolver speed calculation Integrated Circuit (IC) **122**, the proportional scalar IC **124**, the Proportional-Integral-Derivative (PID) IC **126**, and/or the servo drive **127**, can be non-transitory volatile memory (e.g., RAM), non-transitory non-volatile memory (e.g., ROM, flash memory, etc.), or a combination of the two and includes program code for the FPGA **116** as well as a look-up table **128** and a second lookup table **129**.

(35) The first lookup table **128** includes performance data for the rotary fluid device **18**. In one embodiment, and by way of example only, the lookup table **128** includes a relationship between the speed of the shaft **26** of the electric motor **24**, the output pressure of the fluid pump **22** and an output flow of the fluid pump **22**. As the lookup table **128** provides performance characteristics of

the rotary fluid device **18**, the lookup table **128** accounts for performance losses in the pumping assembly **56** and the electric motor **24**. These performance losses include but are not limited to leakage. In certain embodiments, the lookup table **128** can further provide a relationship between phase current supplied to the electric motor **24** and the speed of the shaft **26** and/or a relationship between the phase angle between voltage and current supplied to the electric motor **24** and the torque output of the electric motor **24**. The lookup table **128** can also include temperature variables to account for changes in the relationship between phase current and shaft speed and phase angle and torque due to fluctuations in fluid temperature.

(36) The resolver speed calculation IC **122** calculates the speed of the electric motor **24**. The calculated speed can then be used by the controller **20** to identify a setpoint output pressure of the fluid pump **22** from the lookup table **128** or to identify a setpoint flow of the fluid pump **22** from the lookup table

(37) The proportional scalar IC **124** is used during input control mode to proportionally scale the peak performance values of pressure and/or flow in the lookup table **129** (all other values in the lookup table **129** correspond to those of lookup table **128**). The proportional scalar is only applied to the peak performance values such that if the input command scalar is a 50% command it will result in 50% of full performance (whether it is pressure or flow). In certain embodiments, predetermined peak performances values of pressure and/or flow are stored in a separate memory allocation rather than the lookup table **129**. In this embodiment, the scalar at the input control port **110f**, is used to provide an instruction to the controller **20** to access a certain predetermined peak performance value stored in memory using lookup table **128** and apply the scalar to the predetermined peak performance values. Whether using lookup table **129** or predetermined peak performance values in combination with lookup table **128**, a flow scalar (e.g., 0-10 volts or 0-20 amps) can be provided at the input control port **110f** and/or a pressure scalar (e.g., 0-10 volts or 0-20 amps) can be provided at the input control port **110f** during control mode operation to obtain operation at a desired percentage (e.g. 0-100%) peak performance.

(38) The PID IC **126** compares the setpoint output pressure obtained from the lookup table **128**, **129** to the pressure feedback provided by the pressure transducer **27** and operates to minimize the difference (e.g., error) between the two pressures resulting in an output command signal to the servo drive **127** causing a change in operation of the rotary fluid device **18** to more closely achieve the setpoint output pressure (e.g., the speed of the motor **24** is adjusted, which indirectly adjusts flow due to the pump **22** being a fixed displacement pump, to arrive at the lookup table **128**, **129** pressure value. In certain embodiments, the PID IC **126** is programmed with maximum and minimum command signal limits to ensure the rotary fluid device **18** operates within a desired a range.

(39) The servo-drive **127** receives the command signal from the PID IC **126**, amplifies the signal and transmits electric current to the electric motor **24** in order to produce motion proportional to the command signal to achieve the desired setpoint output pressure.

(40) Referring now to FIGS. **1** and **4**, the operation of the fluid device system **12** in both a normal mode of operation and an input control mode of operation will be described. Voltage is supplied to the circuit **114** of the controller **20** from the power supply **111** (e.g., battery, generator, etc.). With the circuit **114** in a powered state, the FPGA **116** receives sensed operating parameters through the plurality of inputs **110**. The FPGA **116** uses the **116** uses these sensed operating parameters and the lookup table **128** to determine parameters (e.g., voltage, phase current, phase angle, etc.) of the electrical signal **25** that correlate to the desired attribute (e.g., constant horsepower, constant torque, constant speed, pressure compensation, variable pressure compensation, etc.) of the rotary fluid device **18**. The servo drive **127** of the controller **20** outputs the electrical signal **25** having the determined parameters to the electric motor **24**.

(41) In a normal mode of operation, the normal mode input **110d** is set at the electronic switch **113** while a voltage or a current (alternatively, an absence of a voltage or an absence of a current)

provided at the input control port **110f** instructs the controller **20** that the performance of the rotary fluid device **18** is to be 100% of the rated output power, which can be constant pressure control or constant horsepower control per factory settings. FIG. 5 illustrates a pump curve of an electric motor pump whose performance is 100% of rated output power.

(42) By way of example, a generally constant horsepower can be obtained from the pumping assembly **56** by the controller **20** controlling the voltage and current supplied to the electric motor **24** in response to information provided in the lookup table **128**. For example, the horsepower (i.e., HP.sub.motor-in) supplied to the electric motor from the controller **20** can be computed by multiplying the voltage from the controller **20** times the current from the controller **20**. The horsepower out (i.e., HP.sub.motor-out) of the electric motor **24** can be computed by multiplying the horsepower (i.e., HP.sub.motor-in) supplied to the electric motor **24** times the efficiency of the electric motor **24**. In the depicted embodiment, the horsepower out (i.e., HP.sub.motor-out) of the electric motor **24** is generally equal to the horsepower (i.e., HP.sub.pump-in) supplied to the pumping assembly **56**. The horsepower out (i.e., HP.sub.pump-out) of the pumping assembly **56** can be computed by multiplying the horsepower (i.e., HP.sub.pump-in) supplied to the pumping assembly **56** times the efficiency of the pumping assembly **56**. Therefore, the horsepower (i.e., HP.sub.out) out of the rotary fluid device **18** is equal to the voltage supplied by the controller **20** times the current supplied by the controller **20** times the efficiency of the rotary fluid device **18** (i.e., efficiency of the electric motor **24** times the efficiency of the pumping assembly **56**). In one embodiment, the controller **20** receives the efficiency of the rotary fluid device **18** from the lookup table **128** in response to information from at least one of the plurality of inputs **110** of the controller **20**. In another embodiment, the controller computes the efficiency of the rotary fluid device **18** from the information provided by the lookup table **128** based on information from at least one of the plurality of inputs **110** of the controller **20**. Based on this efficiency, the controller **20** can modify, adjust or regulate the voltage, current and phase angle accordingly to maintain a generally constant horsepower from the rotary fluid device **18**.

(43) In an input control mode of operation, the input control mode input **110e** is set at the electronic switch **113** while a desired voltage (e.g., 0-10 volts) or a desired current (0-20 amps) provided at the input control port **110f** as a flow scalar or a pressure scalar. While in the input control mode, an input at the input control port **110f** instructs the controller **20** that the input scalar value should be used to proportionally scale the rated power of the rotary fluid device **18** effectively changing the output pressure or the flow rate of the rotary fluid device **18**. FIG. 6 illustrates an example of a pump curve of an electric motor pump whose performance is 50% of rated output power.

(44) As such, during input control mode operation, the flow and/or pressure scalar is applied to the lookup table peak performance values or to fixed predetermined values stored in memory representing rated performance. The respective value is multiplied by the scalar to determine the new set point value for speed (which is indirectly a new set point value for flow since the pump **22** is a fixed displacement pump) and the pressure sensor **27** signal is also scaled by being multiplied by the scalar to establish a new set point value that is used as the feedback error to adjust the speed.

(45) In an example of input control mode operation, a signal provided at the input control port **110f** can have a full scale of 0 to 10 volts to independently control flow (e.g., speed of the electric motor **24**) or pressure (torque of the electric motor **24**). With a 10 volt signal as the flow scalar at the input control port **110f** the rated speed of the electric motor **24** will be capable of achieving 100% of flow rate depending on an outlet pressure control setting. With a 5 volt signal as the pressure scalar at the input control port **110f** the motor torque of the electric motor **24** will be 50% of rated performance which will produce 50% of the rated pump pressure. With these two settings, the rotary fluid device **18** produces the max rated flow the rotary fluid device **18** was designed for and the output pressure will be 50% reduced from the rated pressure. Utilization of one or both of the flow scalar and the pressure scalar ultimately results in less power consumed by the rotary fluid device **18** allowing a machine or vehicle to effectively manage the power available under changing

load conditions. This feature of utilizing flow and/or pressure scalars is especially beneficial for managing available power in electrical systems that switch between different operational modes to, e.g., a generator mode where power consumption is of a lesser concern and a battery mode where power consumption is of a greater concern. Electrical systems that switch between different operational modes are commonly found in the aerospace industry

(46) Additional uses of the rotary fluid device **18** operating in an input control mode of operation include actuator load and rate control. For actuators requiring rate control, a position feedback signal can be input to the controller **20** of the rotary fluid device **18**. The positional feedback signal changes the electric motor speed (slower to faster) which is directly connected to the hydraulic pump resulting in a closed loop rate control of the actuator enabling accurate speed/rate control of the actuator regardless of aiding or opposing load cases reacting on the actuator.

(47) Regardless of the application, the rotary fluid device **18** operating in an input control mode enables capacity (e.g., fluid volume per unit time) adjustments of a fixed displacement pump, which is otherwise difficult to obtain. While in the input control mode, the controller **20** can also scale the pressure to help manage the overall power consumed.

(48) In either the normal mode of operation or the input control mode of operation of the rotary fluid device **18**, the controller **20**, as opposed to a mechanical spring/spool arrangement, is used as a pressure compensator for the pumping assembly **56** by controlling the voltage and current supplied to the electric motor **24** in response to information provided in the lookup table **128**, **129**. The controller **20** can regulate the outlet pressure from the pumping assembly **56** by regulating the speed of the electric motor **24**, which controls the flow output of the rotary fluid device **18**.

(49) Knowing the speed of the shaft **26** of the rotary fluid device **18** and the current supplied to the electric motor **24**, the controller **20** can determine the torque output of the rotary fluid device **18** by using the lookup table **128**. As torque is a function of pressure and displacement of the rotary fluid device **18** and as the displacement of the rotary fluid device **18** is fixed, the controller **20** can determine the pressure of the rotary fluid device **18** based on this torque determination.

(50) In one embodiment, the controller **20** includes a predefined pressure and/or torque upper limit, which can be scaled according to the scalar when the rotary fluid device **18** is operating in an input control mode. If the controller **20** determines that the pressure or torque output of the rotary fluid device **18** is exceeding this limit, the controller **20** can reduce the pressure or torque by reducing the speed of the electric motor **24**. As the speed of the electric motor **24** decreases, the pressure output from the rotary fluid device **18** also decreases. When the pressure or torque of the rotary fluid device **18** is below the limit, the controller **20** can regulate the speed of the electric motor **24** to maintain the pressure of the rotary fluid device **18**.

(51) In another embodiment, the controller **20** includes the predefined pressure and/or torque upper limit and a lower speed threshold. In this embodiment, if the speed of the electric motor **24** is decreased to the lower speed threshold and the pressure and/or torque of the rotary fluid device **18** has not decreased below the upper limit, the controller **20** stops supplying current to the electric motor **24**. Once the pressure and/or torque of the rotary fluid device **18** falls below the upper limit, the controller **20** will supply current to the electric motor **24**.

(52) As it may be advantageous to control the rotary fluid device **18** as a constant horsepower, constant speed, or constant torque device in a certain applications and in other applications it may be advantageous to control the rotary fluid device **18** as a pressure compensated device. Changing the rotary fluid device **18** to a pressure compensated device can be changed achieved by changing the lookup table **128**. Alternatively, multiple lookup tables **128** can be stored on the storage media and a user selects which lookup table **128** is used by the controller **20** based on the desired operation of the rotary fluid device **18**. For example, the controller **20** may be in electrical communication with a multi-position switch. With the switch in a first position, a first lookup table **128** having performance characteristics for the rotary fluid device **18** in constant horsepower mode is used by the controller **20**. With the switch in a second position, a second lookup table **128** having

performance characteristics for the rotary fluid device **18** in pressure compensation mode is used by the controller **20**. The switch can be manually or electronically operated.

(53) Alternatively, the multiple lookup tables **128** can be selected based on a sensed parameter of the rotary fluid device **18**. For example, the controller **20** can use the first lookup table **128** if the speed of the shaft **26** of the rotary fluid device **18** is above a certain threshold such as 8,000 rpm while the controller **20** can use a second lookup table **128** if the speed of the shaft **26** of the rotary fluid device **18** is below a certain threshold, such as 8,000 rpm. It will be understood, however, that a single lookup table **128** could incorporate the performance characteristics of the first and second lookup tables **128**.

(54) In yet another alternative, the lookup table **128**, which includes the performance characteristics of the rotary fluid device **18**, can be updated. For example, if the rotary fluid device **18** is replaced or if the rotary fluid device **18** is rebuilt, a new lookup table **128** having the performance characteristics of the replacement or rebuilt rotary fluid device **18** can be uploaded or stored on the storage media.

(55) Further details regarding pressure compensation of a fixed displacement electric motor pump can be found in U.S. Pat. No. 10,100,827, which is hereby incorporated by reference.

(56) Various modifications and alterations of this disclosure will become apparent to those skilled in the art without departing from the scope and spirit of this disclosure, and it should be understood that the scope of this disclosure is not to be unduly limited to the illustrative embodiments set forth herein.

Claims

1. A fluid device system comprising: a) a fluid pump having a fluid inlet and a fluid outlet, the fluid pump having a fixed displacement; b) an electric motor having a shaft coupled to the fluid pump, the electric motor being adapted for rotation in response to a power command signal; and c) a controller including an output to communicate the power command signal to the electric motor and including a first input for receiving a first input value, the controller being adapted to adjust the power command signal to operate the electric motor in a normal operating mode and an input control operating mode: i) in the normal operating mode, the controller refers to a lookup table using the first input value and outputs a first power command signal to the electric motor; and ii) in the input control operating mode, the controller calculates a second power command signal based on a scalar value such that the fluid pump is operated at a reduced capacity in comparison to the normal operating mode, wherein the scalar value includes at least one of a pressure scalar value or a flow scalar value, wherein at least one of the flow scalar value or the pressure scalar value is multiplied by a value generated from the lookup table; and wherein the lookup table is a first lookup table and wherein one of the flow scalar value or the pressure scalar is stored in a second lookup table or in memory as a predetermined value.
2. The fluid device system of claim 1, wherein the controller includes a switch adapted to receive an external input for switching the fluid device system between the normal and input control operating modes.
3. The fluid device system of claim 1, wherein the controller includes a second input for receiving at least one of the flow scalar value or the pressure scalar.
4. The fluid device system of claim 3, wherein the second input is configured to receive a variable voltage input or variable current input.
5. The fluid device system of claim 1, wherein the first input value is an output signal from a pressure sensor and wherein the first power command signal is a pressure control signal.
6. The fluid device system of claim 1, wherein the first lookup table provides a correlation between a current supplied to the electric motor and a speed of the electric motor.
7. The fluid device system of claim 1, further comprising at least one sensor in electrical

- communication with the controller, wherein the at least one sensor is adapted to sense a rotational speed of the shaft of the electric motor and a position of the shaft.
8. The fluid device system of claim 1, wherein the scalar value includes both the flow scalar value and the pressure scalar value.
9. The fluid device system of claim 1, wherein the first lookup table is used only in the normal operating mode.
10. The fluid device system of claim 1, wherein the normal operating mode is a constant horsepower mode.
11. The fluid device system of claim 1, wherein one of the pressure scalar value or the flow scalar value is multiplied with a value based on a respective one of a maximum pressure or a maximum flow in the second lookup table.
12. The fluid device system of claim 1, wherein the input control operating mode includes a minimum power command signal and a maximum power command signal.
13. A fluid device system comprising: a) a fluid pump having a fluid inlet and a fluid outlet, the fluid pump having a fixed displacement; b) an electric motor having a shaft coupled to the fluid pump, the electric motor being adapted for rotation in response to a power command signal; and c) a controller including an output to communicate the power command signal to the electric motor and including a first input for receiving a first input value, the controller being adapted to adjust the power command signal to operate the electric motor in a normal operating mode and an input control operating mode: i) in the normal operating mode, the controller refers to a first lookup table using the first input value and outputs a first power command signal to the electric motor; and ii) in the input control operating mode, the controller calculates a second power command signal such that the fluid pump is operated at a reduced capacity in comparison to the normal operating mode; wherein a scalar value proportionally scales a rated power of the fluid pump, the scalar value includes at least one of a pressure scalar value or a flow scalar value, wherein at least one of the flow scalar value or the pressure scalar value is multiplied by a value generated from the first lookup table, and wherein the flow scalar value and/or the pressure scalar value is stored in a second lookup table or in memory as a predetermined value.
14. The fluid device system of claim 13, wherein the controller includes a switch adapted to receive an external input for switching the fluid device system between the normal and input control operating modes.
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